KOS Computing Structures

A comparison using one computing theme

The shared topic is computing/programming. Python appears in DDC, LCC and DBpedia; the other examples concern language categories or broader computer science. These are different levels of the topic, not exact cross-system matches.

Read arrows in their displayed direction. Exact records are in **KOS_Computing_Samples.xlsx**.

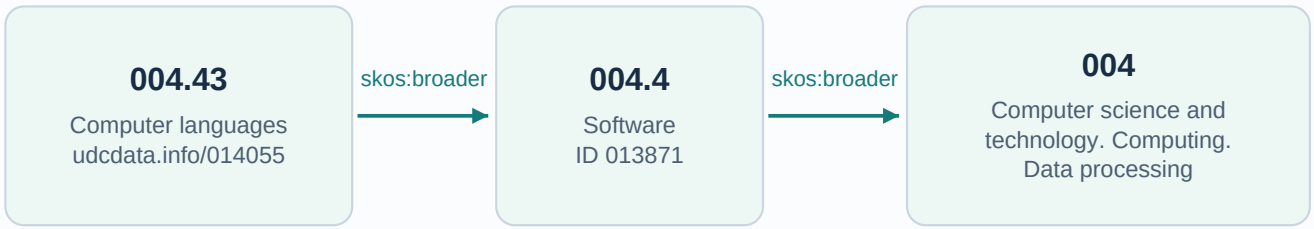
01 DDC: a document and an observed number

A third-party corpus associates this book record with the number 005.

Evidence: companion workbook, DDC and Evidence sheets. dataset_clean.xlsx, row 6303. The assignment is not independently validated; the selected tables provide no official caption or hierarchy.

02 UDC: concepts with broader links

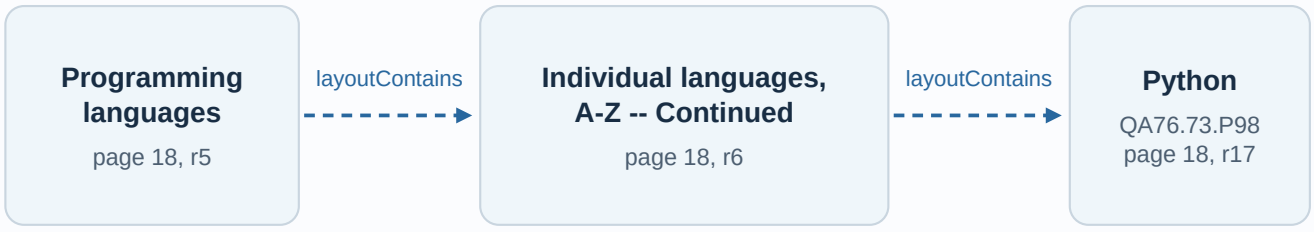
Each arrow leads from a more specific concept to its recorded broader concept.



Evidence: companion workbook, UDC and Evidence sheets. UDC Summary; concept 004 has ID 013566. Statements rows 3388 and 3358 preserve the native direction.

03 LCC: the context of a printed schedule entry

Python occurs under a continued list of individual programming languages. The dashed edges were derived from page indentation.



Evidence: companion workbook, LCC and Evidence sheets. LCC_Q2024TEXT.pdf, page 18; local entry keys end in r5, r6 and r17. QA76.73.P98 is the contextual notation derived for the Python entry.

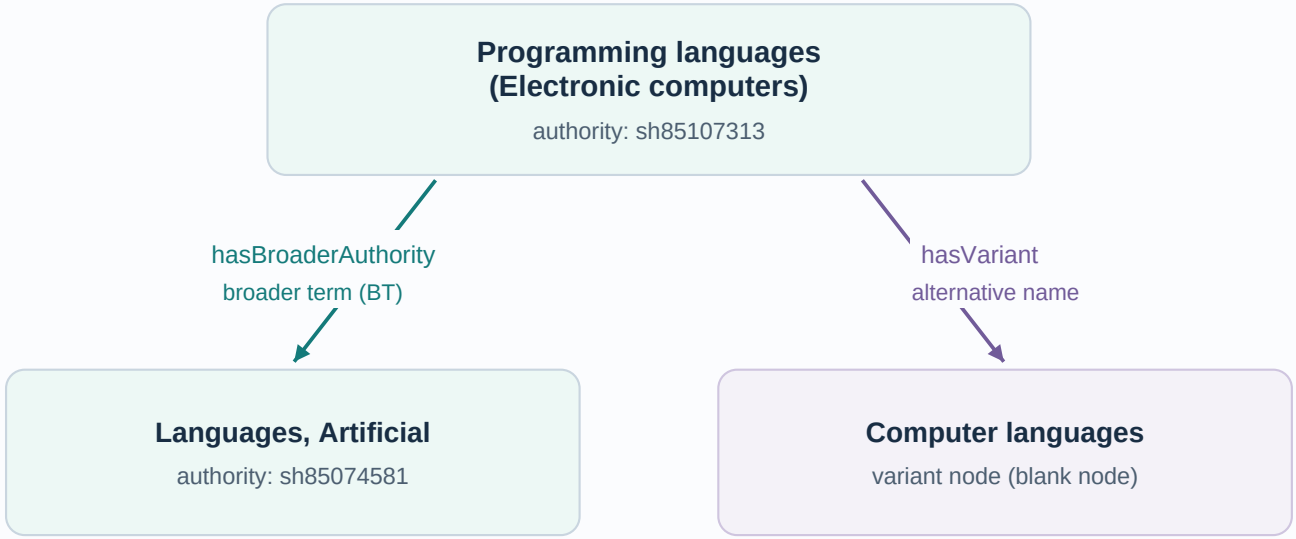
Read these as schedule-layout links. They do not assert that Python is a subclass of a list heading.

Headings and Formal Classes

A subject vocabulary and an ontology organize the topic differently

04 LCSH: an authority, a broader term and a variant

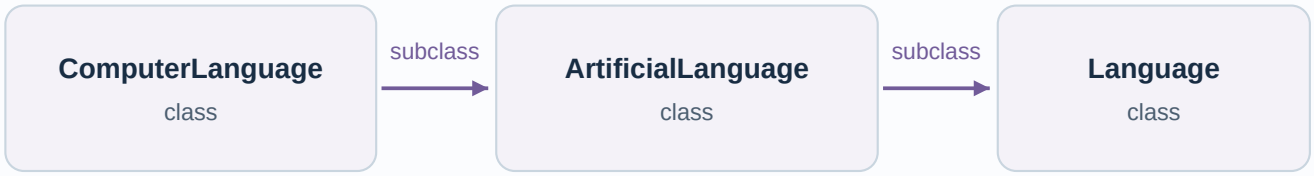
The preferred heading has a broader authority and an alternative name. The alternative name is stored on a separate variant node.



Evidence: companion workbook, LCSH and Evidence sheets. LC MADS/RDF authority sh85107313; native arrow directions are retained.

05 SUMO: class axioms and a general rule

Named classes are connected by subclass axioms. Formal rules also state what follows from class membership.



Native general rule

```
(=> (and (subclass ?X ?Y)
         (instance ?Z ?X))
    (instance ?Z ?Y))
```

If X is a subclass of Y and Z is an instance of X, then Z is an instance of Y. The symbols ?X, ?Y and ?Z are variables; no named individual is supplied by this example.

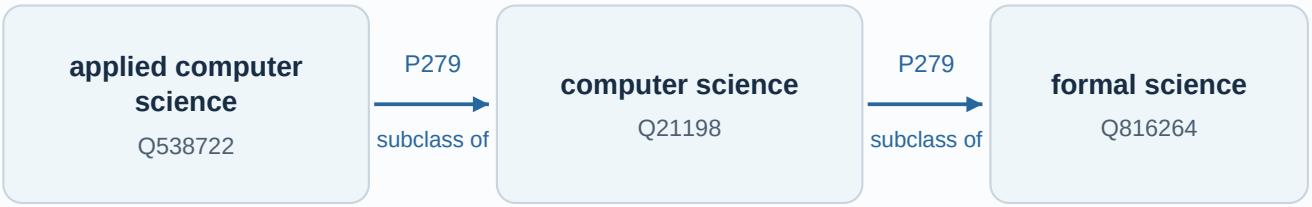
Evidence: companion workbook, SUMO and Evidence sheets. Merge.kif. Rule SUMOEX:Merge.kif:000025 appears in SUMO_Prepared_Native_Logic.xlsx, Native_Logic row 26. Line breaks here are for readability.

Statements and Typed Resources

The selected edges retain the meanings used in each source

06 Wikidata: statements about computer science

This example uses broader computer science, because the downloaded records do not supply a retrieved programming-language example.

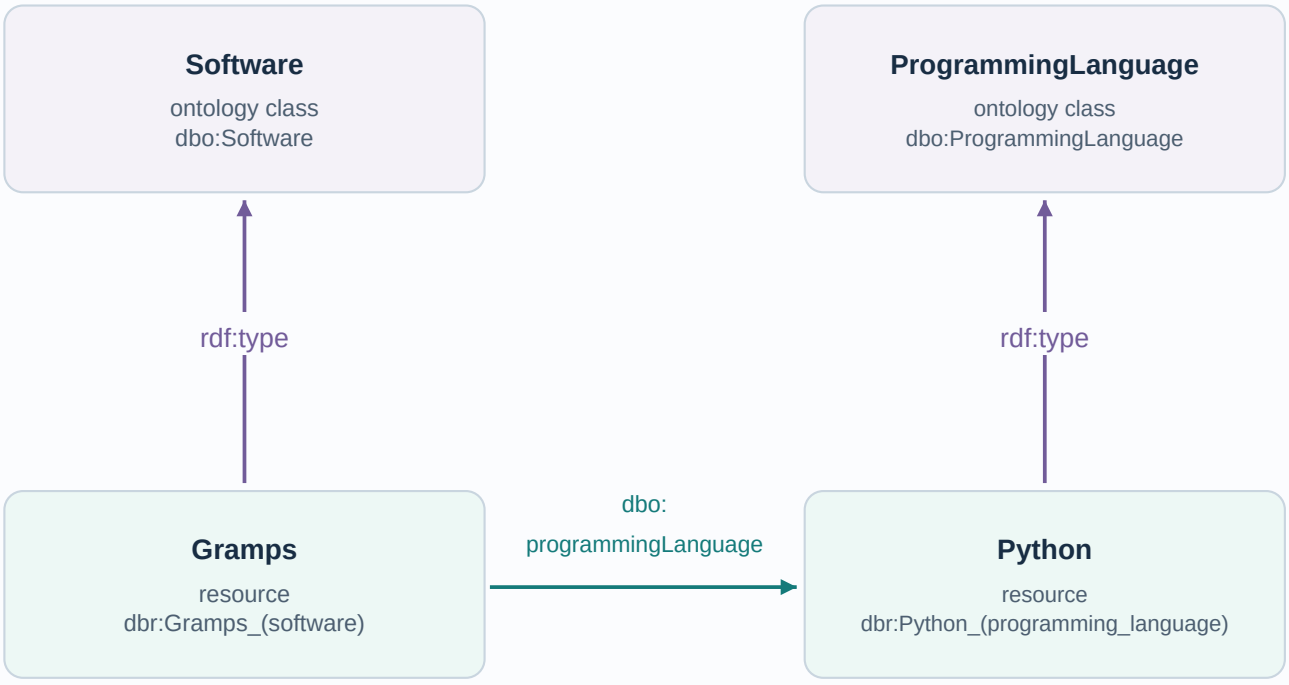


Both retained P279 statements have normal rank. Neither has a retained qualifier; the arrows show the main statement values only.

Evidence: companion workbook, Wikidata and Evidence sheets. Statements_001 rows 72803 and 8710; source entity IDs are shown in the nodes.

07 DBpedia: resource facts and class membership

Gramps is linked to Python by a programming-language fact. Separate type statements connect each resource to an ontology class.



Prefixes: dbo: = dbpedia.org/ontology/; dbr: = dbpedia.org/resource/.

Evidence: companion workbook, DBpedia and Evidence sheets. DBpedia_Prepared.xlsx, Statements rows 881, 3103 and 3583. Resources and ontology classes are distinct node roles.